

Windows Privilege Escalation & Persistence - Incident Report

Date of Report: 20/09/2025

Date of Incident: 19/09/2025

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1. Preparation

Incident Title: Privilege Escalation & Persistence on DESKTOP-5LV1F3O

Date/Time: September 19, 2025 (02:16 – 02:29)

Affected Host: DESKTOP-5LV1F3O (192.168.1.105)

Service Targeted: Local Windows accounts, services, and WSL installation

Scope: Review Wazuh logs related to group membership changes, privilege assignments, service creation, and subsystem installation.

Tools Used: Wazuh log monitoring via EventChannel.

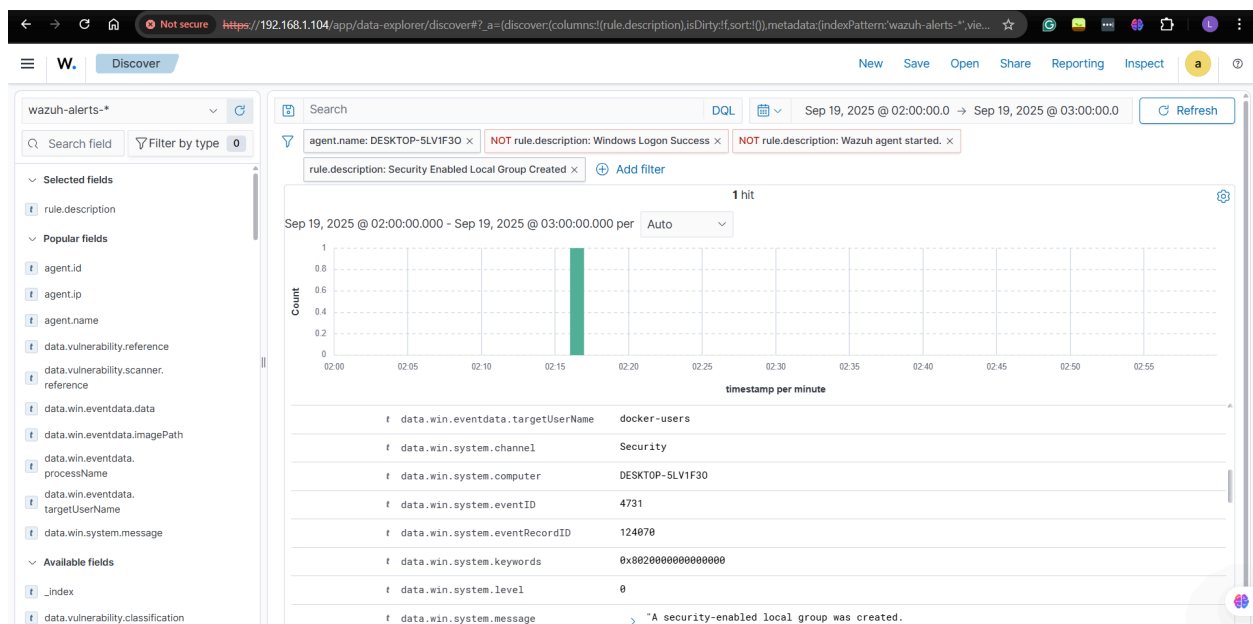
2. Detection & Analysis

2.1 Victim Machine (DESKTOP-5LV1F3O)

Timeline of Events

02:16:48 | Local Group Manipulation

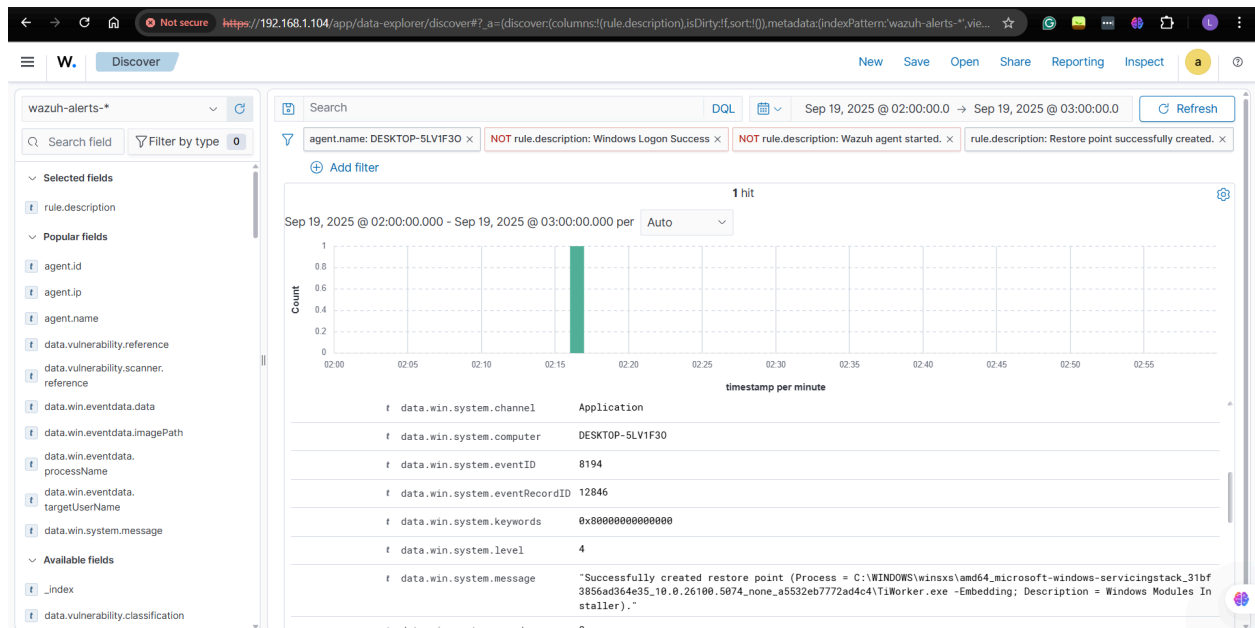
- Multiple events indicate the creation and modification of new local groups.
- The user Dev (SID: S-1-5-21-...-1002) was added to the docker-users group.



[Reference](#)

02:16:59 | Restore Point Creation

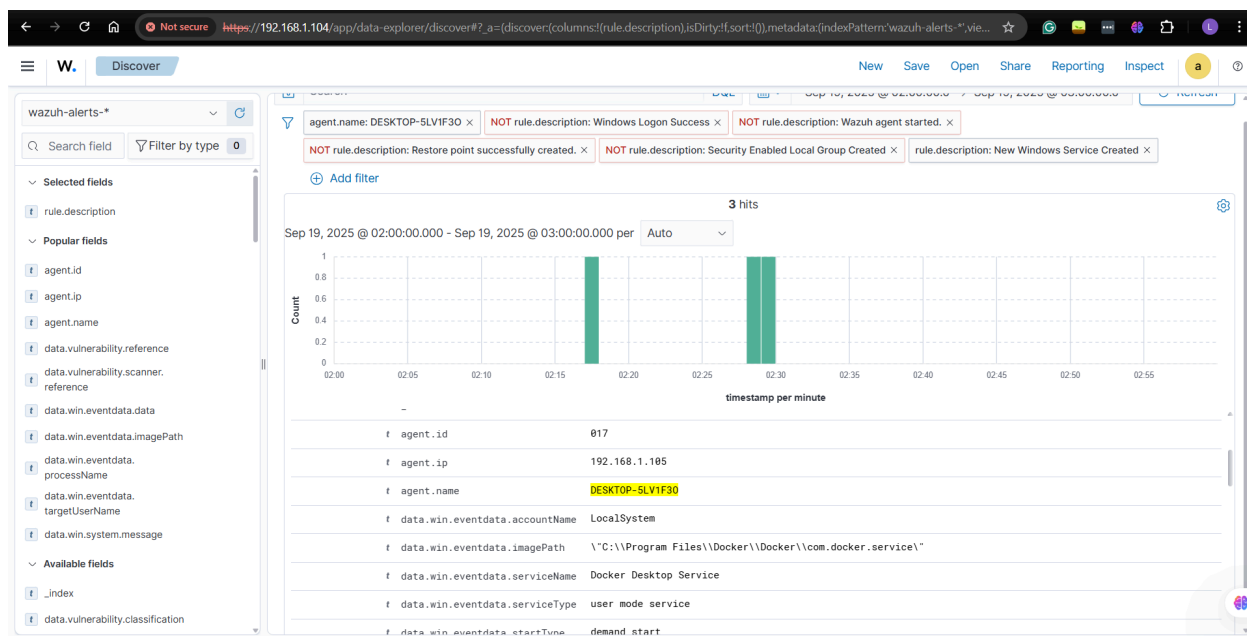
- A Windows restore point was created immediately after group modifications.
- Suggests preparation for rollback in case attacker actions break functionality.



[Reference](#)

02:17:04 | New Windows Service Created

- A suspicious Windows service was created.
- New service creation is a common Persistence technique (MITRE T1543).



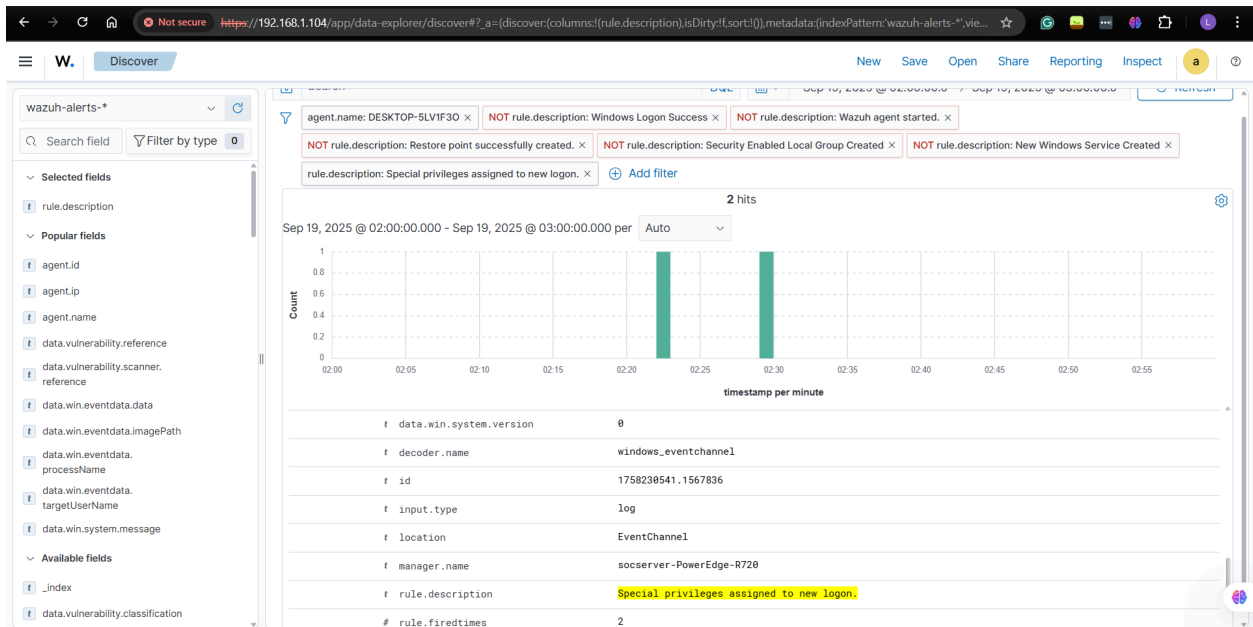
[Reference](#)

02:20:46 – 02:20:48 | System Restart & Logoff

- System shutdown initiated, followed by user logoff.
- Multiple system errors logged during shutdown.
- Likely intended to apply privilege/group/service changes.

02:22:21 | Suspicious Logon with Special Privileges

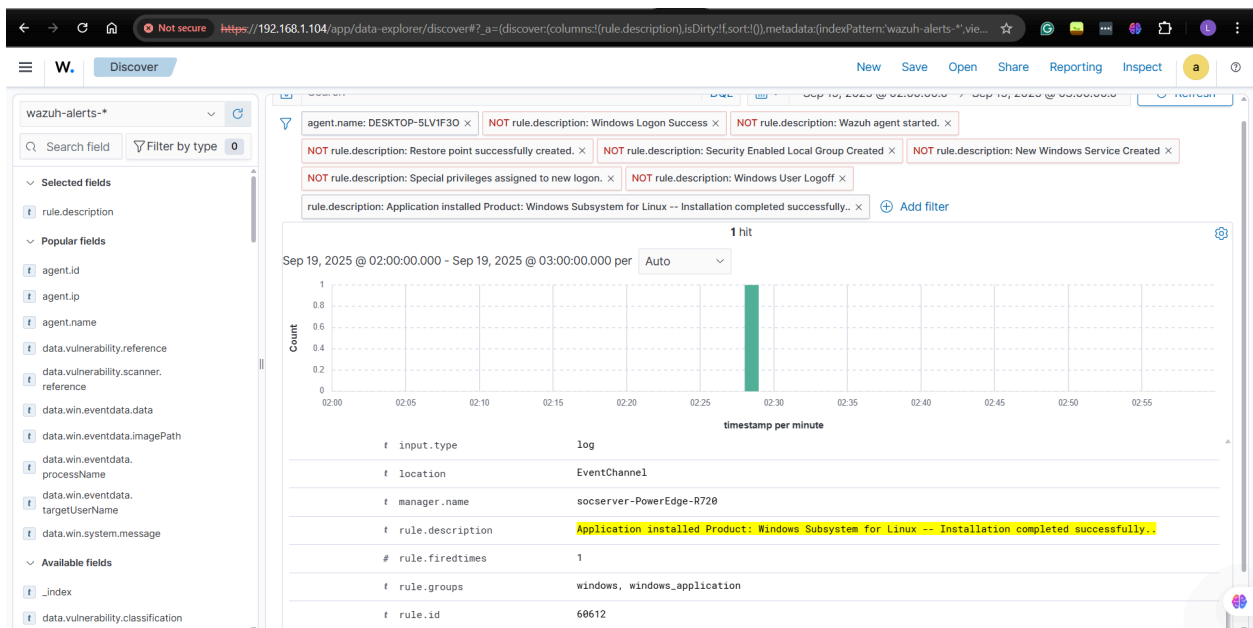
- Workstation logon success recorded.
- Special privileges are assigned to the session.
- Suggests attacker re-entered the host with elevated rights.



Reference

02:28:00 – 02:28:09 | Windows Subsystem for Linux (WSL) Installed

- Windows Installer launched.
- WSL v2.6.1.0 installed successfully.
- WSL enables Linux tools (such as nmap, netcat, and tcpdump) to run under Windows, often bypassing endpoint monitoring. This is a high-risk persistence & execution vector.



[Reference](#)

02:28:09 – 02:29:20 | Additional Services & Privilege Assignments

- Multiple new Windows services created.
- A service logon from a non-standard service account was recorded.
- Special privileges are again assigned to a logon.
- Analysis: Confirms attacker persistence attempts and ongoing privilege escalation.

2.2 Key Observations

- Activity began with local group creation and privilege escalation at 02:16.
- A restore point was created (unusual, attacker preparation).
- New services were deployed for persistence.
- System reboot applied changes.
- WSL is installed to enable Linux-based attack tools on Windows.
- Multiple privileged logons and service logons followed.

3. Analysis of the Attack Technique

Privilege Escalation: Local group manipulation, granting Dev elevated roles (MITRE T1484).

Persistence: Creation of new Windows services and service accounts (MITRE T1543).

Defense Evasion: Use of WSL to run Linux binaries inside Windows (MITRE T1059.004).

Execution: Installation of WSL tools enabling further network or credential attacks.

Preparation: A Restore point was created to recover.

4. Executive Summary

On September 19, 2025, the workstation DESKTOP-5LV1F3O showed clear evidence of malicious activity consistent with post-compromise attacker behavior.

The attack began at 02:16 with suspicious local group modifications, giving the account Dev elevated rights. A restore point was immediately created, likely as a failsafe. Soon after, new services were deployed, ensuring persistence.

The system was then restarted, and a privileged logon was observed at 02:22. At 02:28, the attacker installed Windows Subsystem for Linux (WSL), enabling execution of Linux-based tools under Windows. Finally, additional suspicious services and privileged logons were recorded.

This sequence confirms privilege escalation, persistence establishment, and preparation for further malicious operations.

5. Detection and Response Recommendations

5.1 Detection

Enable Wazuh rules for:

- Group creation/modification (EventID 4731/4732/4735).
- New service creation (EventID 7045).
- WSL installation and MSI installer events.
- Logons with special privileges.

5.2 Response

- Immediately audit the Dev account and groups (docker-users).
- Review and remove unauthorized services created around 02:16–02:29.
- Uninstall or investigate the WSL installation.
- Collect forensic artifacts (Event Logs, Prefetch, Service Registry Keys).
- Reset all credentials for local accounts.
- Consider reimaging the host if root-level persistence is confirmed.

6. Lessons Learned

- Privilege escalation via group modification went undetected until post-factum analysis.
- Attackers leveraged legitimate features (Restore Points, WSL) for malicious purposes.
- Service creation and privileged logons are critical red flags requiring real-time alerting.
- Endpoint hardening is required: restrict WSL installation and service creation to admins only.
- Stronger monitoring of group membership changes is needed to prevent privilege escalation abuse.

